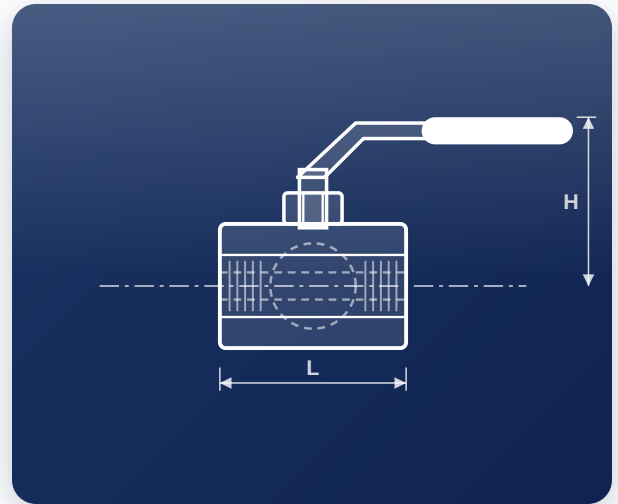


PRODUCT CATALOG

Instrumentation Ball Valves

Quarter-turn shut-off and switching valves for instrumentation, sampling, hydraulic and process lines — from **1/8" tube to 2" pipe**, rated up to **10,000 psi**.

This catalog is written to help you pick the right valve quickly, check its dimensions and order it correctly the first time. Every series page shows what the valve is for, what it's made of and the sizes available.


THE RANGE AT A GLANCE

 SIZES 1/8" – 2" Plus metric tube ends from 4 mm to 25 mm OD	 PRESSURE 10,000 psi Three classes: 3,000 · 6,000 · 10,000 psi	 CONNECTIONS NPT · BSP BSPT, ISO, socket weld, SAE and twin-ferrule tube ends	 MATERIALS 316 SS + 316L/304/304L SS, brass, carbon steel, Monel, Inconel, Hastelloy, duplex
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WHAT'S INSIDE

—	How to choose the right ball valve	02	05	Round body two-piece, 3,000 psi	13
—	Materials, ratings & part numbers	03	06	Hex body two-piece, 6,000 psi	15
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03	Instrument angle, 6,000 psi	09	09	Heavy-duty three-piece, 10,000 psi	19
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Not sure which series fits? Start on **page 02** — or call us and describe your application.

HOW TO CHOOSE THE RIGHT BALL VALVE

Four questions narrow this catalog down to **one series** — answer them in order.



STEP 1 · PRESSURE

How much pressure will the valve see?

Pick a rating above your highest system pressure, including surges. **3,000 psi** round-body valves suit gauges and general utility lines; the **6,000 psi** series cover most instrumentation; the **10,000 psi** series are built for hydraulics and high-pressure oil & gas. Remember the seat can limit the rating (page 03).



STEP 2 · FLOW PATH

What should the flow do?

Two-way valves simply open and close a line. An **angle** valve does the same while turning the line 90°, saving you an elbow fitting. A **3-way** valve diverts or switches flow between two lines with 180° of handle travel. 5-way switching versions are available on request.



STEP 3 · END CONNECTION

How will it connect to your system?

Female **NPT**, **BSP** or **BSPT** threads are the most common. **Twin-ferrule tube ends** (fractional or metric) connect straight to tubing with no thread sealant — fewer leak paths and no sealant contamination. Socket weld and SAE ports are available on selected series.



STEP 4 · MATERIAL

What fluid will it carry?

Match the body and seat material to the fluid and environment. 316 stainless steel is the safe default; the material guide on the next page explains when to choose something else.

SERIES AT A GLANCE

#	SERIES	RATING	SIZES	ENDS	PAGE
01	Instrument, one-piece body	6,000 psi	1/8" – 1"	FNPT · tube	04
02	Instrument, three-piece	6,000 psi	1/8" – 1" · 4–25 mm	FNPT · tube	06
03	Instrument angle	6,000 psi	1/4" – 1"	FNPT/BSP · tube	09
04	Instrument 3-way	6,000 psi	1/4" – 1"	FNPT/BSP · tube	11
05	Round body, two-piece	3,000 psi	1/4" – 2"	FNPT	13
06	Hex body, two-piece	6,000 psi	1/4" – 1"	FNPT	15
07	Square body, three-piece	6,000 psi	1/4" – 1"	FNPT	16
08	Hex body, high pressure	10,000 psi	1/4" – 1/2" · 6–12 mm	FNPT · MNPT · tube	17
09	Heavy-duty three-piece	10,000 psi	1/8" – 1"	NPT · SAE	19

MATERIALS, RATINGS & PART NUMBERS

Choose the material for the **fluid**, then check the **seat** — it often sets the real pressure limit.

MATERIAL GUIDE — INSTRUMENT SERIES

MATERIAL	A GOOD CHOICE FOR	BODY CWP	WITH PTFE SEATS
316 Stainless Steel	The default: water, air, gases, oils and most chemicals	6,000 psig (414 bar)	1,500 psig (103 bar)
Brass	Water, air and non-corrosive service at lower cost	3,000 psig (207 bar)	1,500 psig (103 bar)
Monel Alloy 400	Seawater, brines and chloride-rich media	3,000 psig (207 bar)	1,500 psig (103 bar)
Hastelloy C-276	Aggressive acids and chemicals where stainless pits	4,000 psig (276 bar)	1,500 psig (103 bar)

SEATS SET THE LIMIT

A valve is only as strong as its weakest part. With **PTFE seats** these instrument valves are limited to **1,500 psig**, even when the body is rated higher. Also available: 304/304L SS, carbon steel, Inconel and duplex. Tell us your pressure, temperature and fluid and we'll confirm the right seat.

READING A PART NUMBER

TYA PREFIX	BV BALL VALVE	2 FLOW PATTERN	S STAINLESS	04 SIZE	FF ENDS
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Flow pattern: 2 = two-way · 2V / 2A = angle · 3V = three-way. **Size:** sixteenths of an inch (02 = 1/8", 04 = 1/4", 08 = 1/2", 16 = 1"); an **M** prefix means metric tube (M08 = 8 mm OD). **Ends:** FF = female thread both ends · OD = tube ends.

Series 05–07 use the body shape (**RO** round, **HX** hex, **SQ** square) with sizes 4 = 1/4", 6 = 3/8", 8 = 1/2", 12 = 3/4", 25 = 1", 38 = 1-1/2", 50 = 2". Series 08 uses TE / MTE for inch / metric tube ends, MF for male NPT and FF for female NPT.

BUILT INTO EVERY SERIES



Floating ball

Self-adjusts against the seat to make up for wear



Blow-out-proof stem

Stem is loaded from inside so pressure cannot eject it



Micro-finished ball

Smooth surface for a positive, bubble-tight seal



Adjustable packing

Stem seal can be tightened without removing the valve



Quarter-turn action

Fast, low-torque operation with a positive stop

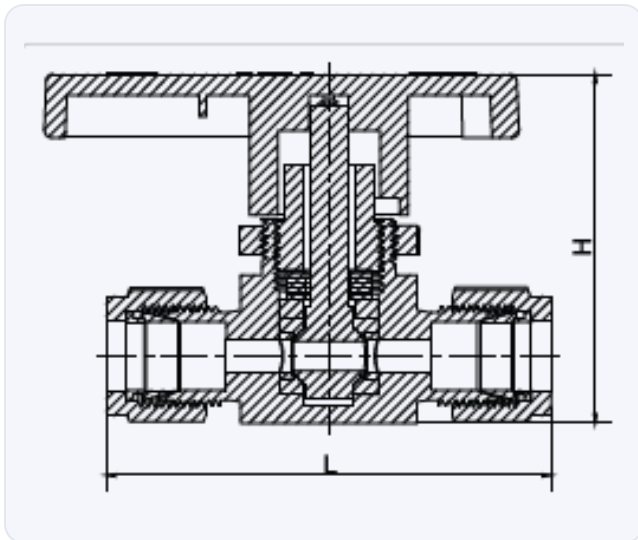


Flow indication

Handle position shows flow direction at a glance

ONE-PIECE BODY INSTRUMENT VALVE

Up to 6,000 psi · 1/8" – 1" · female NPT or tube ends · panel mountable



A compact valve with a single-piece body — no body joints that could leak — and a very small internal volume, so little fluid is left trapped to contaminate the next sample.

- ✓ Blow-out-proof, bottom-loaded stem
- ✓ Low-torque nylon handle with brass insert
- ✓ Full or reduced bore; PTFE, POM or Delrin seats
- ✓ Panel nut included for mounting through a panel

🎯 **Best for:** analyzer and sampling systems that need the lowest dead volume, and control panels that need a compact, high-flow valve with clear flow indication.

SIZES & DIMENSIONS

PART NUMBER	SIZE (TUBE OD / FNPT)	L · FNPT	L1 · TUBE	H	ORIFICE
TYA-BV-2-S-02-FF / OD	1/8"	51	51	51	3
TYA-BV-2-S-04-FF / OD	1/4"	56	56	56	5
TYA-BV-2-S-06-FF / OD	3/8"	63	66	65	8
TYA-BV-2-S-08-FF / OD	1/2"	90	99	99	10
TYA-BV-2-S-12-FF / OD	3/4"	90	99	99	16
TYA-BV-2-S-16-FF / OD	1"	115	121	112	19

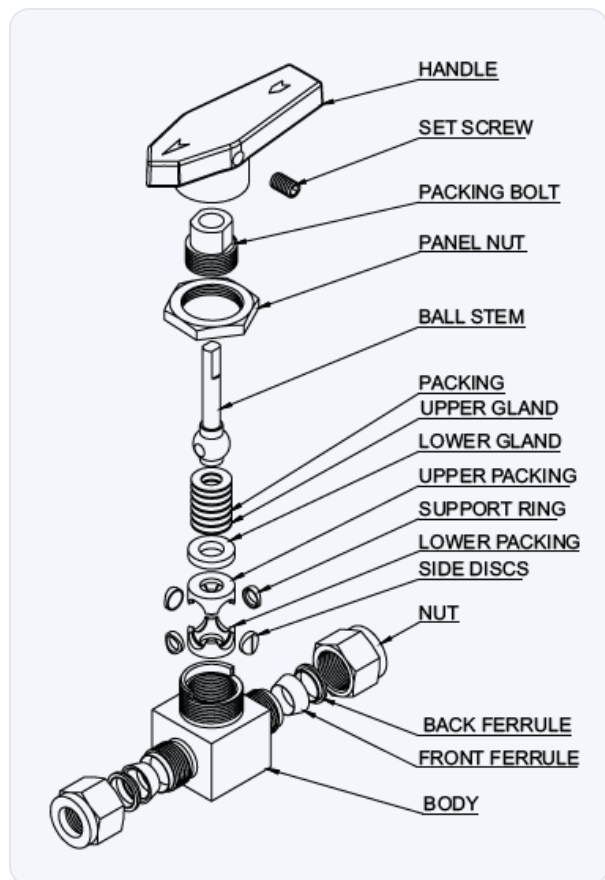
L = female NPT length · L1 = tube-end length · H = overall height · mm, for reference only.

PRESSURE RATING BY MATERIAL

MATERIAL	BODY CWP	WITH PTFE SEATS
316 Stainless Steel	6,000 psig (414 bar)	1,500 psig (103 bar)
Brass	1,000 psig (70 bar)	1,000 psig (70 bar)
Monel Alloy 400	3,000 psig (207 bar)	1,500 psig (103 bar)
Hastelloy C-276	4,000 psig (276 bar)	1,500 psig (103 bar)

ONE-PIECE BODY — HOW IT GOES TOGETHER

Every part, **top to bottom**, in the same order as the exploded view



SERVICING TIP

A slight weep at the stem can usually be cured in line by snugging the **packing bolt (03)** a small amount. Always depressurise the line before loosening any other part.

 **Panel mounting:** remove the panel nut (04), pass the stem through the panel hole and refit the nut from the front.

01	Handle	Nylon, brass insert	×1
	Quarter-turn grip; its position shows open/closed and flow direction.		
02	Set screw	Steel	×1
	Locks the handle onto the stem.		
03	Packing bolt	A479-316L	×1
	Compresses the stem packing — snug it to stop a stem weep.		
04	Panel nut	A479-316L	×1
	Clamps the valve through a panel for front-of-panel mounting.		
05	Ball stem	A479-316L	×1
	One-piece ball and stem; bottom-loaded so it cannot blow out.		
06	Packing	A479-316L	×6
	Ring stack that keeps steady load on the stem seals.		
07	Upper gland	A479-316L	×1
	Guides the stem and transfers load to the packing.		
08	Lower gland	A479-316L	×1
	Seats the packing stack around the stem.		
09	Upper packing	PTFE	×1
	Primary stem seal.		
10	Support ring	A479-316L	×2
	Backs up the PTFE packing so it cannot extrude.		
11	Lower packing	PTFE	×1
	Secondary stem seal.		
12	Side discs	A479-316L	×2
	Discs either side of the ball that locate and seal it.		
13	Nut	A479-316L	×2
	Tube-end nut; tightening it makes up the ferrules.		
14	Back ferrule	A479-316L	×2
	Grips the tube against pressure and vibration.		
15	Front ferrule	A479-316L	×2
	Seals between the tube and the valve body.		
16	Body	A479-316L	×1
	Single-piece body with no body joints to leak.		

*Nut and ferrules are fitted to tube-end versions. **A479-316L** = 316L stainless bar.*

THREE-PIECE INSTRUMENT VALVE

6,000 psi · 1/8" – 1" female NPT · 1/8" – 1" and 4 – 25 mm tube ends



A square body with two O-ring-sealed end adaptors and a free-floating ball. The straight-through bore keeps pressure drop low, and a 90° turn takes it from fully open to fully closed. Ideal for instrument, impulse and sample lines.

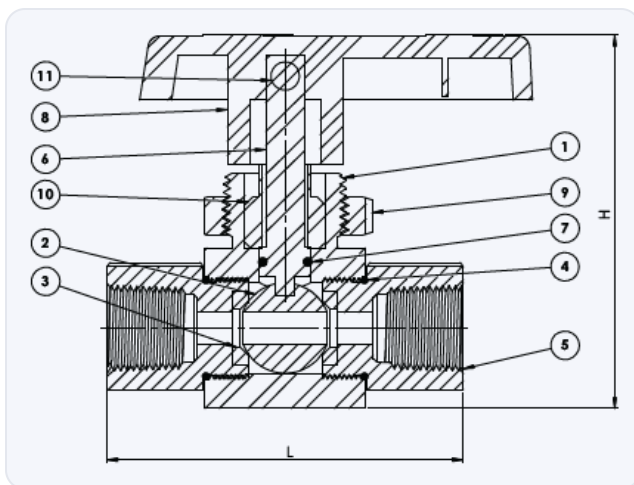
- ✓ Free-floating ball compensates for seat wear
- ✓ Micro-finished ball for a positive seal
- ✓ 90° actuation with phenolic handle or SS lever
- ✓ Adjustable PTFE stem seal, serviceable in line

FEMALE NPT × FEMALE NPT

PART NUMBER	SIZE	L	H	ORIFICE
TYA-BV-2-S-02-FF	1/8" female NPT	70	50	3
TYA-BV-2-S-04-FF	1/4" female NPT	70	50	5
TYA-BV-2-S-06-FF	3/8" female NPT	72	50	8
TYA-BV-2-S-08-FF	1/2" female NPT	85	60	10
TYA-BV-2-S-12-FF	3/4" female NPT	98	60	16
TYA-BV-2-S-16-FF	1" female NPT	115	70	19

L overall length · H overall height · mm, reference only · **B-16** = brass option · parts below shared by Series 02–04.

CROSS-SECTION · COMMON PARTS, SERIES 02–04



01	Body	A479-316L/B-16	x1
02	Ball	A479-316	x1
03	Seal	PTFE/POM	x2
04	O-ring	Viton/Buna-N	x2
05	Adaptor	A479-316L/B-16	x2
06	Spindle	A479-316	x1
07	O-ring	Viton/Buna-N	x1
08	Handle	Phenolic / SS lever	x1
09	Panel nut	A479-316L/B-16	x1
10	Gland nut	A479-316L/B-16	x1
11	Grub screw	Steel	x1

THREE-PIECE VALVE — TUBE ENDS

6,000 psi · twin-ferrule tube ends in 1/8" – 1" and 4 – 25 mm OD



The same three-piece valve with twin-ferrule tube ends built in, so it connects straight to your tubing. There are no taper threads and no thread sealant in the system — fewer leak paths, no sealant contamination and a neater, faster installation.

- ✓ Free-floating ball compensates for seat wear
- ✓ Micro-finished ball for a positive seal
- ✓ Straight-through flow path, minimum pressure drop
- ✓ 90° actuation; adjustable PTFE stem seal in line

FRACTIONAL TUBE OD

PART NUMBER	TUBE OD	L	H	ORIF.
TYA-BV-2-S-02-OD	1/8" OD	78	50	3
TYA-BV-2-S-04-OD	1/4" OD	78	50	5
TYA-BV-2-S-05-OD	5/16" OD	78	50	5
TYA-BV-2-S-06-OD	3/8" OD	85	50	8
TYA-BV-2-S-08-OD	1/2" OD	102	60	10
TYA-BV-2-S-10-OD	5/8" OD	97	63	10
TYA-BV-2-S-12-OD	3/4" OD	115	60	16
TYA-BV-2-S-16-OD	1" OD	121	73	19

METRIC TUBE OD

PART NUMBER	TUBE OD	L	H	ORIF.
TYA-BV-2-S-M04-OD	4 mm OD	78	50	3
TYA-BV-2-S-M06-OD	6 mm OD	78	50	5
TYA-BV-2-S-M08-OD	8 mm OD	78	50	5
TYA-BV-2-S-M10-OD	10 mm OD	85	50	8
TYA-BV-2-S-M12-OD	12 mm OD	102	60	10
TYA-BV-2-S-M16-OD	16 mm OD	97	63	10
TYA-BV-2-S-M20-OD	20 mm OD	115	60	16
TYA-BV-2-S-M25-OD	25 mm OD	121	73	19

L overall length · H overall height · Dimensions in mm, for reference only — request a certified drawing before finalising a design.

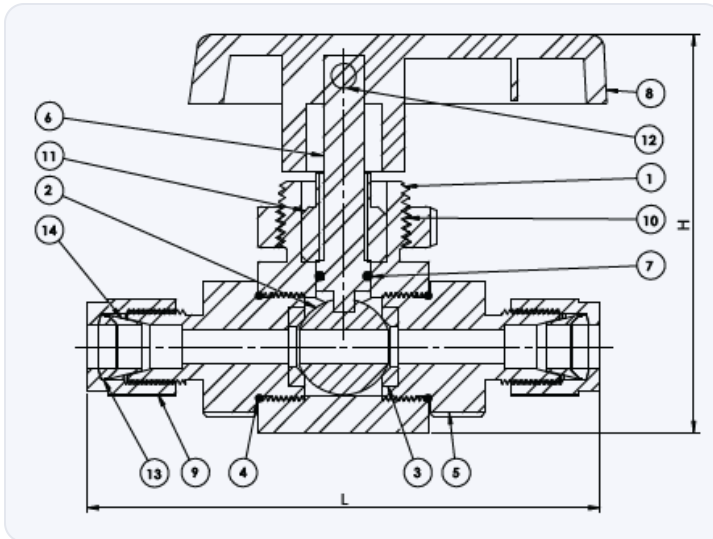


MATCH THE TUBE OD

Tube ends must match your tubing's **outside diameter** exactly. Some inch and metric sizes look almost identical — 6 mm and 1/4", 10 mm and 3/8", 12 mm and 1/2" — but they are not interchangeable. Metric part numbers carry an **M** before the size, and nuts or ferrules must never be mixed between sizes, standards or manufacturers.

TUBE END — CROSS-SECTION & PARTS

Numbers in the list match the **callouts on the drawing**



01	Body	x1
02	Ball	x1
03	Seal	x2
04	O-ring	x2
05	Adaptor	x2
06	Spindle	x1
07	O-ring	x1
08	Handle	x1
09	Nut	x2
10	Panel nut	x1
11	Gland nut	x1
12	Grub screw	x1
13	Back ferrule	x2
14	Front ferrule	x2



MATERIALS — SEE PAGE 06

Same materials as the common parts list.
Tube nut (09) and ferrules (13, 14): A479-316L/B-16.

HOW A TWIN-FERRULE TUBE END SEALS



STEP 1 Prepare & insert

Cut the tube square and deburr it. Push it through the nut (09) and ferrules until it bottoms firmly on the shoulder inside the fitting.



STEP 2 Tighten

As the nut is tightened the front ferrule (14) seals against the fitting body and tube, while the back ferrule (13) grips the tube to resist pressure and vibration.



STEP 3 Remake when needed

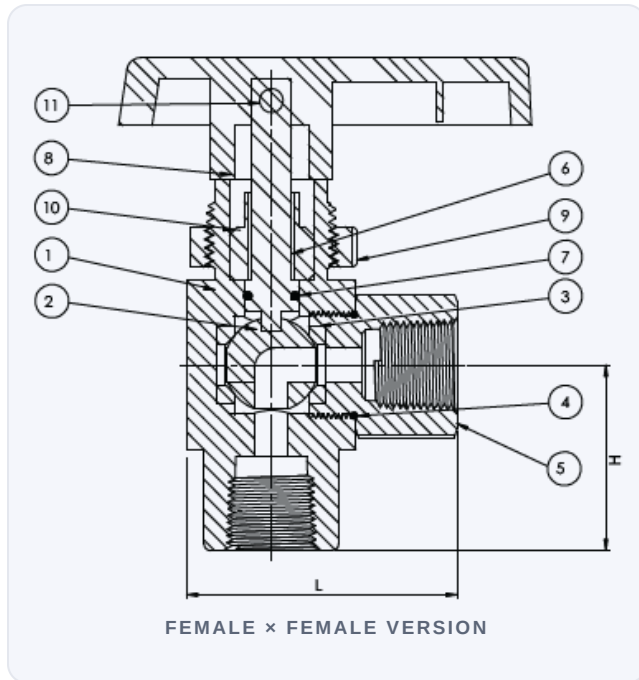
The joint can be disconnected for maintenance and retightened. Keep each ferrule set with its tube end — never reuse ferrules on a different fitting.



Servicing: a slight weep at the stem can usually be cured in line by snugging the gland nut (11). Depressurise the line before loosening any tube nut.

ANGLE PATTERN INSTRUMENT VALVE

6,000 psi · 1/4" – 1" · female NPT/BSP or tube ends · 90° flow path



Shuts off flow and turns the line through 90° in one component. Using an angle valve instead of a straight valve plus an elbow saves a joint, a potential leak path and panel space.

- ✓ Free-floating ball compensates for seat wear
- ✓ Micro-finished ball for a positive seal
- ✓ Positive handle stop — no over-travel
- ✓ Adjustable PTFE stem seal, serviceable in line

PARTS — NUMBERS MATCH THE DRAWING

01	Body	x1	02	Ball	x1
03	Seal	x2	04	O-ring	x2
05	Adaptor	x2	06	Spindle	x1
07	O-ring	x1	08	Handle	x1
09	Panel nut	x1	10	Gland nut	x1
11	Grub screw	x1			

FEMALE × FEMALE (NPT/BSP)

PART NUMBER	SIZE (NPT/BSP)	L	H	ORIFICE
TYA-BV-2V-S-04-FF	1/4" female	41	32	3
TYA-BV-2V-S-06-FF	3/8" female	41	32	8
TYA-BV-2V-S-08-FF	1/2" female	45	38	10
TYA-BV-2V-S-12-FF	3/4" female	55	58	16
TYA-BV-2V-S-16-FF	1" female	55	58	19

L and H as shown on the drawing. Dimensions in mm, for reference only — request a certified drawing before finalising a design. Tube-end versions are on the [next page](#).



SAME PARTS AS SERIES 02 — MATERIALS ON PAGE 06

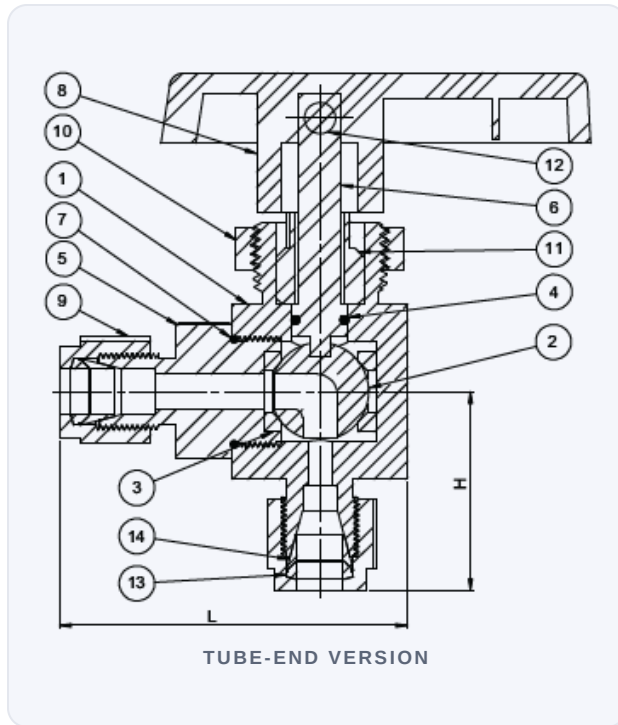
Items 01–11 are identical to the three-piece valve, in the same order, so one materials list covers both.



Best for: tight panel layouts, gauge and transmitter connections, and anywhere the line must change direction at the valve.

ANGLE VALVE — TUBE ENDS

6,000 psi · 1/4" – 1" tube OD · twin-ferrule ends · 90° flow path



The same angle valve with twin-ferrule tube ends on both ports, so it connects straight to tubing and turns the line 90° — no elbow, no taper threads and no thread sealant in the system.

- ✓ Free-floating ball compensates for seat wear
- ✓ Micro-finished ball for a positive seal
- ✓ Positive handle stop — no over-travel
- ✓ Adjustable PTFE stem seal, serviceable in line

PARTS — NUMBERS MATCH THE DRAWING

01 Body	×1	02 Ball	×1
03 Seal	×2	04 O-ring	×1
05 Adaptor	×1	06 Spindle	×1
07 O-ring	×1	08 Handle	×1
09 Nut	×2	10 Panel nut	×1
11 Gland nut	×1	12 Grub screw	×1
13 Front ferrule	×2	14 Back ferrule	×2

TUBE ENDS

PART NUMBER	TUBE OD	L	H	ORIFICE
TYA-BV-2A-04-OD	1/4"	41	35	5
TYA-BV-2A-06-OD	3/8"	41	35	8
TYA-BV-2A-08-OD	1/2"	45	40	10
TYA-BV-2A-12-OD	3/4"	55	60	16
TYA-BV-2A-16-OD	1"	55	60	19

L and H as on the drawing · mm, reference only · tube OD must match your tubing — see [page 07](#).



MATERIALS ON PAGE 06

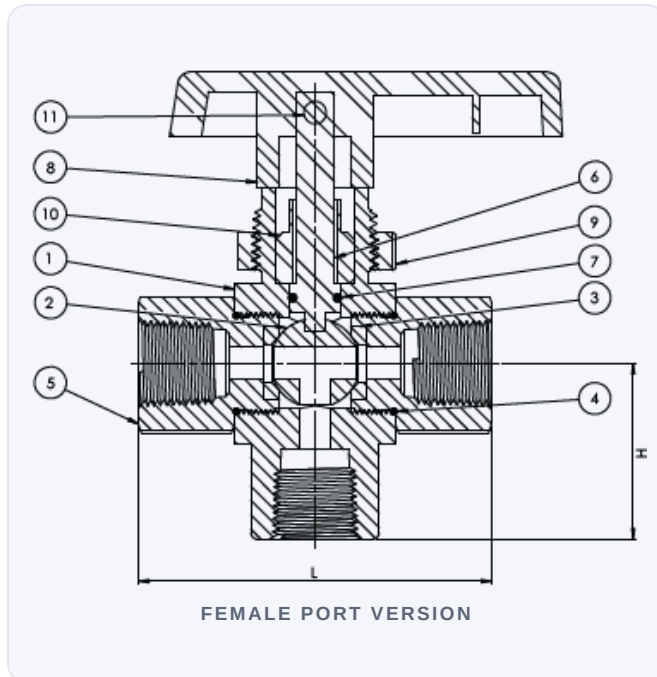
Nut (09) and ferrules (13, 14): A479-316L/B-16; all other parts as the common list.



Best for: tubed panels and analyzers needing a compact, sealant-free change of direction.

3-WAY SWITCHING INSTRUMENT VALVE

6,000 psi · 1/4" – 1" · female NPT/BSP or tube ends · 180° actuation



One valve that switches flow between two lines — for example, directing a sample to an analyzer or to a vent, or selecting between two supply sources. The handle points the way the fluid is going.

- ✓ Diverts or selects flow with 180° of handle travel
- ✓ Handle indicates flow direction
- ✓ Free-floating ball compensates for seat wear
- ✓ Micro-finished ball; adjustable PTFE stem seal

PARTS — NUMBERS MATCH THE DRAWING

01	Body	×1	02	Ball	×1
03	Seal	×2	04	O-ring	×2
05	Adaptor	×2	06	Spindle	×1
07	O-ring	×1	08	Handle	×1
09	Panel nut	×1	10	Gland nut	×1
11	Grub screw	×1			

FEMALE PORTS (NPT/BSP)

PART NUMBER	SIZE (NPT/BSP)	L	H	ORIFICE
TYA-BV-3V-S-04-FF	1/4" female	70	32	3
TYA-BV-3V-S-06-FF	3/8" female	72	32	8
TYA-BV-3V-S-08-FF	1/2" female	85	38	10
TYA-BV-3V-S-12-FF	3/4" female	98	58	16
TYA-BV-3V-S-16-FF	1" female	115	58	19

L and H as shown on the drawing. Dimensions in mm, for reference only — request a certified drawing before finalising a design. Tube-end versions are on the **next page**.



SAME PARTS AS SERIES 02 — MATERIALS ON PAGE 06

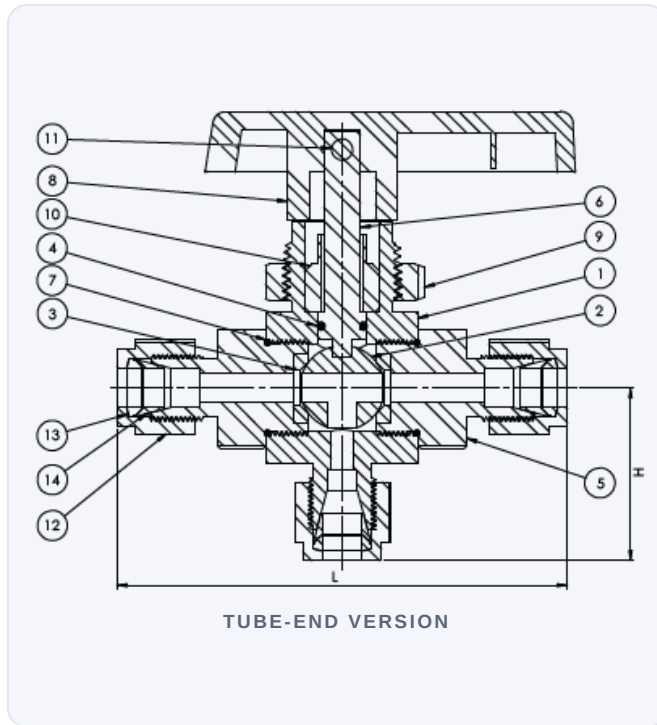
Items 01–11 are identical to the three-piece valve, in the same order, so one materials list covers both.



Best for: stream switching, dual-source selection and vent diversion — one valve instead of two.

3-WAY VALVE — TUBE ENDS

6,000 psi · 1/4" – 1" tube OD · twin-ferrule ends on all three ports · 180° actuation



The same 3-way switching valve with twin-ferrule tube ends on all three ports, so it connects straight to tubing — no taper threads and no thread sealant in the sample path.

- ✓ Diverts or selects flow with 180° of handle travel
- ✓ Handle indicates flow direction
- ✓ Free-floating ball compensates for seat wear
- ✓ Micro-finished ball; adjustable PTFE stem seal

PARTS — NUMBERS MATCH THE DRAWING

01	Body	×1	02	Ball	×1
03	Seal	×2	04	O-ring	×1
05	Adaptor	×2	06	Spindle	×1
07	O-ring	×2	08	Handle	×1
09	Panel nut	×1	10	Gland nut	×1
11	Grub screw	×1	12	Nut	×3
13	Back ferrule	×3	14	Front ferrule	×3

TUBE ENDS

PART NUMBER	TUBE OD	L	H	ORIFICE
TYA-BV-3V-S-04-OD	1/4"	78	35	5
TYA-BV-3V-S-06-OD	3/8"	85	35	8
TYA-BV-3V-S-08-OD	1/2"	102	40	10
TYA-BV-3V-S-12-OD	3/4"	115	60	16
TYA-BV-3V-S-16-OD	1"	121	60	19

L and H as on the drawing · mm · tube OD must match — see [page 07](#).



MATERIALS ON PAGE 06

Nut (12) and ferrules (13, 14): A479-316L/B-16.



Best for: tubed analyzer and sampling systems that switch streams without threaded joints.



ROUND BODY TWO-PIECE BALL VALVE

3,000 psi · 1/4" – 2" female NPT · the widest size range in this catalog



A two-piece, bi-directional valve with a lever handle for quick quarter-turn operation. It is the practical choice for gauge isolation and general utility service, and it is the only series here that goes up to 2".

- ✓ Bubble-tight shut-off with PTFE or PEEK ball seats
- ✓ Heat code traceable to a certified material test report
- ✓ Anti-blow-out stem, low operating torque
- ✓ Quarter-turn handle with positive stop; every valve factory tested

SIZES & DIMENSIONS

PART NUMBER	SIZE	L	D	S	H	A	B
TYA-RO2P-4-FF	1/4"	55	27	23	78.5	53.5	37.5
TYA-RO2P-6-FF	3/8"	55	27	23	78.5	53.5	37.5
TYA-RO2P-8-FF	1/2"	73	35	30	106	69.5	49
TYA-RO2P-12-FF	3/4"	78.5	44.5	38	157	67.5	45
TYA-RO2P-25-FF	1"	90	55.5	43	156	81.5	49.8
TYA-RO2P-38-FF	1-1/2"	131.5	92	75	235	155	101.5
TYA-RO2P-50-FF	2"	136.5	97	80	240	160	106.5

*L length · D body diameter · S across flats · H handle reach · A overall height · B centreline to handle top · mm. See the drawing on the **next page**.*

RATING · SEAT · BALL

3,000 psi · soft seat · hard-chrome ball

BODY MATERIALS

316L/316/304/304L SS, Monel, Inconel, carbon steel, duplex · NACE MR0175

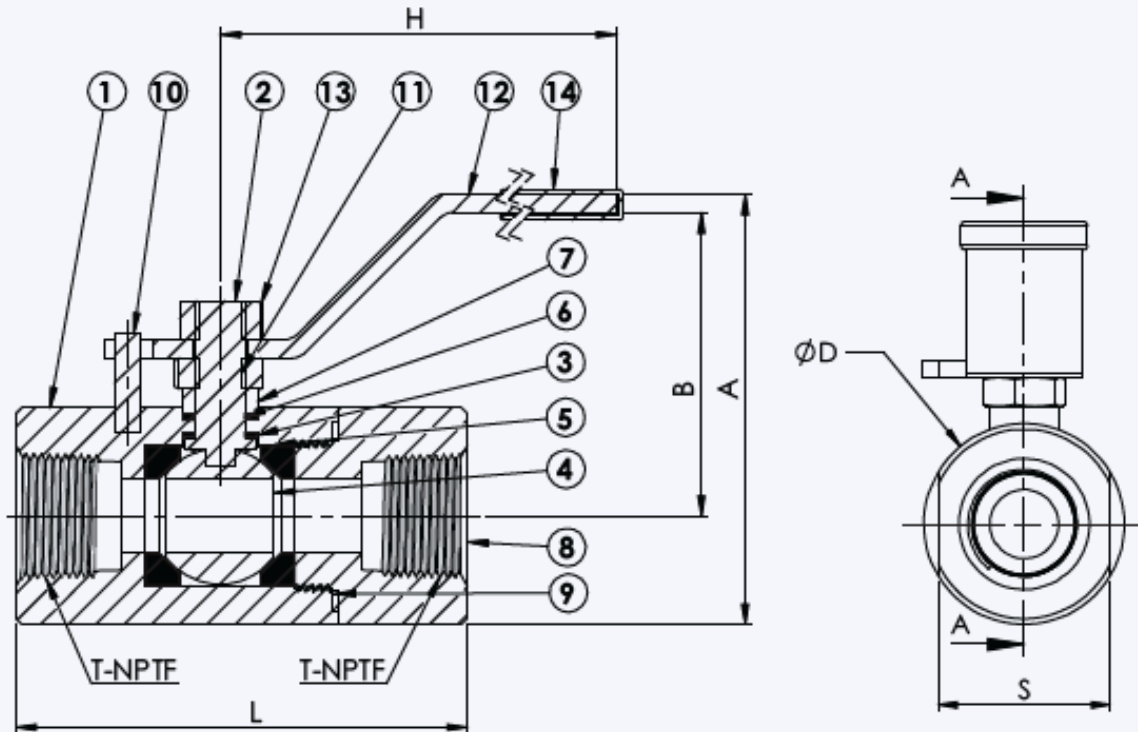
SERVICE · CONNECTIONS

Liquid, gas, vapour · NPT, BSP, BSPT, ISO, socket weld, tube end

🎯 **Best for:** isolating pressure gauges and instruments, plus water, air and oil utility lines up to 2".

ROUND BODY — CROSS-SECTION & PARTS

Dimension letters match the size table on **page 13**; part numbers match the **callouts**



WHAT IT'S MADE OF — COMMON TO SERIES 05, 06 & 07

01	Body	A479-316L/A105	×1	08	Adaptor	A479-316L/A105	×1
02	Stem	A276-316	×1	09	Body seal	PTFE/Graphoil	×1
03	Stem packing	PTFE/Graphoil	×1	10	Lock pin	A479-316L/A105	×1
04	Ball	A479-316L/A105	×1	11	Gland nut	A479-316L/A105	×1
05	Ball seal	PTFE/Graphoil	×2	12	Handle	A276-316L/A105	×1
06	Gland packing	PTFE/Graphoil	as req.	13	Lock nut	A479-316L/A105	×1
07	Gland washer	A276-316L/A105	as req.	14	Sleeve	PVC	×1

Material codes: **A479 / A276-316(L)** = 316/316L stainless bar · **B-16** = brass alternative · **A105** = carbon steel alternative.



SERVICING

A slight weep at the stem can usually be cured in line by snugging the gland nut (11). Depressurise the line before any other disassembly.

HEX BODY TWO-PIECE BALL VALVE

6,000 psi (414 bar) • 1/4" – 1" female NPT • hex body for easy wrenching



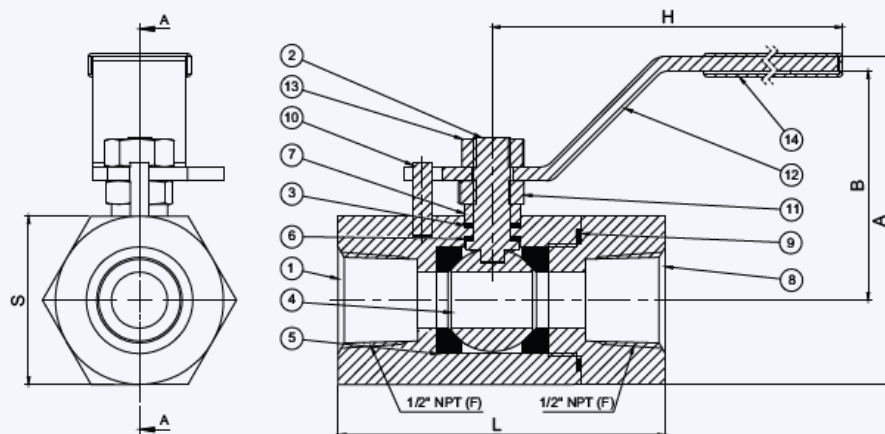
Twice the pressure class of the round body in a hexagonal body you can hold securely with a spanner while making up threaded connections — so the torque goes into the joint, not the valve. Ideal for higher-pressure instrument, hydraulic and process lines.

- ✓ Free-floating ball compensates for seat wear
- ✓ Micro-finished, hard-chrome-plated ball
- ✓ Straight-through flow path for minimum pressure drop
- ✓ Adjustable PTFE stem seal, serviceable in line

SIZES & DIMENSIONS

PART NUMBER	SIZE	L	S	H	A	B
TYA-HX2P-4-FF	1/4"	55	27	78.5	53.5	37.5
TYA-HX2P-6-FF	3/8"	55	27	78.5	53.5	37.5
TYA-HX2P-8-FF	1/2"	73	36	106	69.5	49
TYA-HX2P-12-FF	3/4"	78.5	45	157	67.5	45
TYA-HX2P-25-FF	1"	90	55	156	81.5	49.8

L length • *S* across flats • *H* handle reach • *A* overall height • *B* centreline to handle top • F×F NPT • mm



 **Same as Series 05:** callouts 1–14 match the parts list on page 14; specs as page 13, rated 6,000 psi.

SQUARE BODY THREE-PIECE BALL VALVE

6,000 psi (414 bar) · 1/4" – 1" female NPT · robust, compact and easy to operate



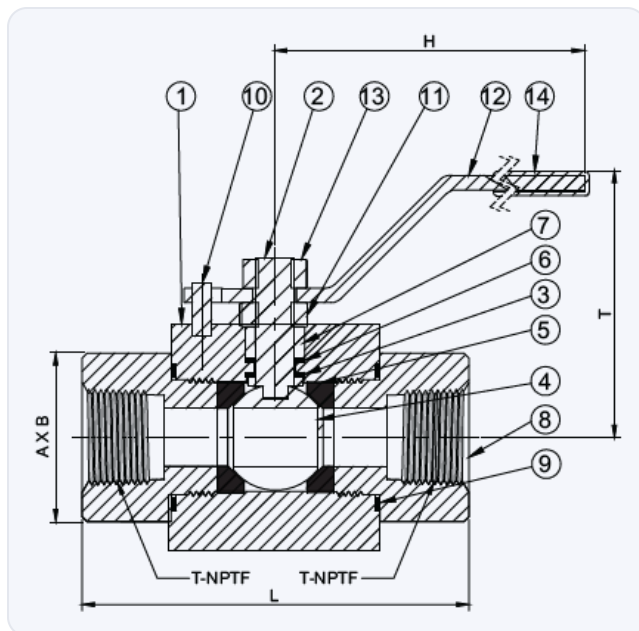
An internally loaded, blow-out-proof stem and a floating ball give positive shut-off in a compact square body. The hard-chromed ball reduces friction, so the valve stays easy to turn throughout its life.

- ☑ Hard-chromed ball for low friction and low torque
- ☑ Blow-out-proof, internally loaded stem
- ☑ PTFE and Graphoil gland packings
- ☑ Adjustable stem seal; other materials and ends on request

SIZES & DIMENSIONS

PART NUMBER	SIZE	A	B	L	H	T	ORIFICE
TYA-SQ2P-4-FF	1/4"	30	33	76	110	49	4.8
TYA-SQ2P-6-FF	3/8"	30	33	76	110	49	7
TYA-SQ2P-8-FF	1/2"	35	38	96	180	64	9.5
TYA-SQ2P-12-FF	3/4"	43	49	108	180	67	12
TYA-SQ2P-25-FF	1"	50	56	125	180	70	20

All sizes female NPT both ends (F×F), mm, for reference only. Tube, socket-weld and other ends on request.



READING THE DRAWING

- A × B** Square body section
- L** Overall length
- H** Handle reach from stem centre
- T** Bore centreline to top of handle
- T-NPTF** Female NPT port thread

 **Same as Series 05:** callouts 1–14 match the parts list on page 14; specs as page 13, rated 6,000 psi.

 **Best for:** demanding process and utility service where you want a rugged, flat-sided body that mounts and wrenches easily.

HEX BODY HIGH-PRESSURE BALL VALVE

10,000 psi (689 bar) · 1/4" – 1/2" NPT · 1/4" – 1/2" and 6 – 12 mm tube ends



A true two-piece, bi-directional design that keeps body leak paths to a minimum. Integral tube ends remove taper threads and sealant altogether — less contamination and fewer leak paths.

- ✓ Bubble-tight shut-off with PEEK ball seats
- ✓ Full hydrostatic and low-pressure air tested
- ✓ Anti-blow-out stem; PTFE gland packing
- ✓ Quarter-turn handle with protective sleeve

SIZES & DIMENSIONS

PART NUMBER	SIZE	A	B	C-HEX	D-HEX
TYA-BLVHX-SS-6MTE	6 mm OD	95	166	22	30
TYA-BLVHX-SS-8MTE	8 mm OD	97	166	32	38
TYA-BLVHX-SS-10MTE	10 mm OD	100	168	32	38
TYA-BLVHX-SS-12MTE	12 mm OD	105	170	36	41
TYA-BLVHX-SS-4TE	1/4" OD	95	166	22	30
TYA-BLVHX-SS-6TE	3/8" OD	99	168	32	38
TYA-BLVHX-SS-8TE	1/2" OD	105	170	36	41
TYA-BLVHX-SS-4MF	1/4" male NPT	70	162	22	30
TYA-BLVHX-SS-8MF	1/2" male NPT	85	167	36	41
TYA-BLVHX-SS-4FF	1/4" female NPT	70	163	22	30
TYA-BLVHX-SS-6FF	3/8" female NPT	71	162	32	38
TYA-BLVHX-SS-8FF	1/2" female NPT	85	167	36	41

A body length · B overall length incl. handle · C / D hex sizes · letters as on the **page 18** drawing · mm

RATING · SEAT · BALL

10,000 psi (689 bar) · PEEK soft seat · hard-chrome ball

BODY MATERIALS

316L/316/304/304L SS, Monel, Inconel, carbon steel, duplex

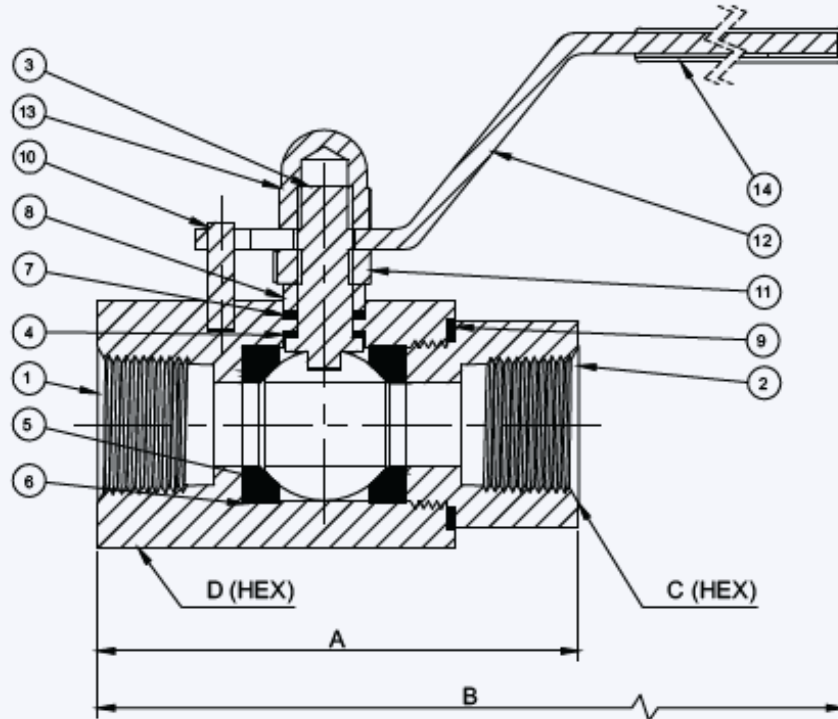
SERVICE · CONNECTIONS

Liquid, gas, vapour · NPT, BSP, BSPT, ISO, socket weld, tube end

⊙ **Best for:** the most demanding oil, gas and process control applications, including hydraulic and high-pressure gas lines.

10,000 PSI HEX — CROSS-SECTION & PARTS

Dimension letters match the size table on **page 17**; part numbers match the **callouts**



WHAT IT'S MADE OF — NUMBERS MATCH THE DRAWING

01	Body	A479-316/A105	x1	08	Gland washer	A276-316/A105	x1
02	Adaptor	A479-316/A105	x1	09	Body seal	PTFE	x1
03	Stem	A276-316	x1	10	Lock key	A479-316/A105	x1
04	Stem packing	PTFE	x1	11	Gland nut	A479-316/A105	x1
05	Ball	ASTM A182 F316	x1	12	Handle	SS 304	x1
06	Ball seal	PEEK	x2	13	Lock nut	A479-316/A105	x1
07	Gland packing	PTFE	x1	14	Sleeve	PVC, moulded	x1

Material codes: **A479 / A276-316** = 316 stainless bar · **A182 F316** = forged 316 stainless · **A105** = carbon steel alternative · **PEEK** = high-strength polymer seat.

SERVICING

A slight weep at the stem can usually be cured in line by snugging the gland nut (11). At 10,000 psi, always depressurise and vent the line fully before loosening anything.

HEAVY-DUTY THREE-PIECE BALL VALVE

10,000 psi (689 bar) · 1/8" – 1" · NPT or SAE ports · built for hydraulics



A simple, heavy-duty way to control hydraulic flow: a ported ball held between seals in a solid square housing. It needs no lubrication or routine maintenance, and every valve can be fully repaired.

- ☑ Fully bidirectional — cannot be installed the wrong way round
- ☑ No threads in the fluid path to shed debris
- ☑ Hard-chrome, micro-smooth ball cuts friction and seat wear
- ☑ Completely repairable; no lubrication needed for life

PORTS & LENGTH (MM)

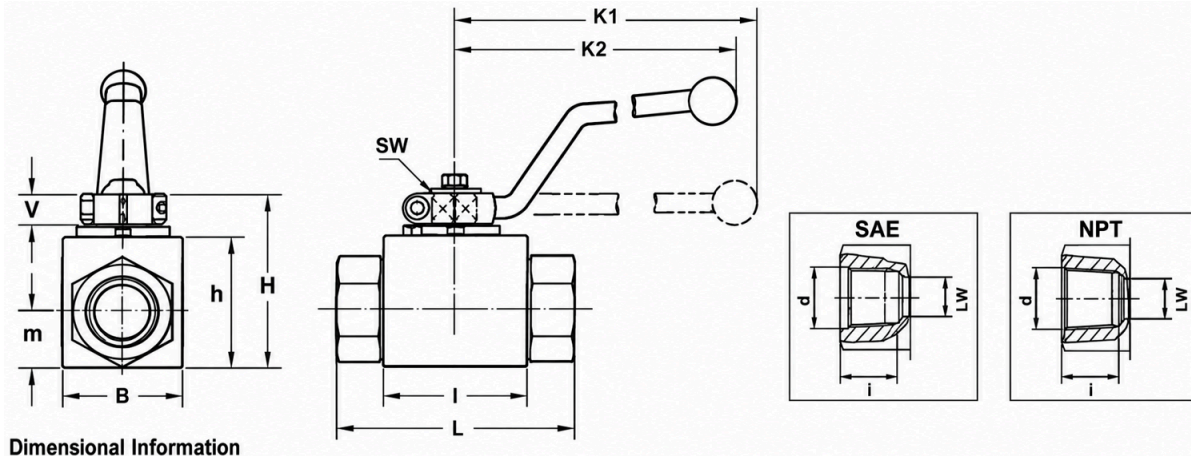
SIZE	d · NPT	d · SAE	i · NPT	i · SAE	L	LW · NPT	LW · SAE
1/8"	1/8–27	5/16–24	12.95	—	—	6.10	—
1/4"	1/4–18	7/16–20	17.02	13.97	69.00	6.10	5.08
3/8"	3/8–18	9/16–18	17.53	15.49	72.00	10.00	10.00
1/2"	1/2–14	3/4–16	15.80	17.53	83.06	13.00	13.00
3/4"	3/4–14	1-1/16–12	18.00	23.11	95.00	20.00	20.00
1"	1–11-1/2	1-5/16–12	27.69	20.00	113.00	25.00	25.00
1" H	1-1/2–11-1/2	1-5/8–12	20.00	20.00	113.00	25.00	25.00

BODY & HANDLE (MM)

SIZE	I	B	H	h	m	V	SW	K1	K2
1/8"	39.88	25.91	46.99	33.02	13.46	10.92	9	150.11	115.06
1/4"	39.88	25.91	46.99	33.02	13.46	10.92	9	150.11	115.06
3/8"	42.93	32.00	52.07	38.10	17.53	10.92	9	150.11	115.06
1/2"	48.01	35.05	54.10	39.88	19.05	10.92	9	150.11	115.06
3/4"	61.98	49.02	74.93	56.90	24.38	13.97	14	199.90	171.00
1"	66.04	57.91	83.06	65.02	29.46	13.97	14	199.90	160.00
1" H	74.00	70.00	88.00	70.00	34.50	14.00	14	200.00	160.00

Every letter is explained on the dimension drawing on the **next page**. Values in mm, for reference only.

HEAVY-DUTY VALVE — DIMENSIONAL INFORMATION

Drawing and dimension chart in **inches and mm** · materials: 316/304 SS, Monel, Inconel, carbon steel, duplex

Dimensional Information

Size	d NPT	d SAE	i NPT	i SAE	L SAE / NPT	LW NPT	LW SAE	I	B	H	h	m	V	SW	K1	K2
1/8"	1/8 27	5/16 24	0.51 12.95	0.55 13.97	2.72 69.00	0.24 8.10	0.2 5.08	1.57 39.88	1.02 25.91	1.85 46.99	1.30 33.02	0.53 13.46	0.43 10.92	0.35 9	5.91 150.11	4.53 115.08
1/4"	1/4 18	7/16 20	0.67 17.02	0.61 15.49	2.83 72.00	0.39 10.00	0.39 10.00	1.69 42.93	1.26 32.00	2.05 52.07	1.50 38.10	0.69 17.53	0.43 10.92	0.35 9	5.91 150.11	4.53 115.08
3/8"	3/8 18	9/16 18	0.69 17.53	0.69 17.53	3.27 83.06	0.51 13.00	0.51 13.00	1.89 48.01	1.38 35.05	2.13 54.10	1.57 39.88	0.75 19.05	0.43 10.92	0.35 9	5.91 150.11	4.53 115.08
1/2"	1/2 14	3/4 16	0.62 15.80	0.71 18.00	3.74 95.00	0.79 20.00	0.79 20.00	2.44 61.98	1.93 49.02	2.95 74.93	2.24 56.90	0.96 24.38	0.55 13.97	0.55 14	7.87 199.90	6.73 171
3/4"	3/4 14	1 1/16 12	0.71 18.00	0.79 20.00	4.45 113.00	0.98 25.00	0.98 25.00	2.60 66.04	2.28 57.91	3.27 83.08	2.56 65.02	1.16 29.46	0.55 13.97	0.55 14	7.87 199.90	6.30 160
1"	1 11 1/2	1 5/16 12	1.09 27.69	0.79 20.00	4.45 113.00	0.98 25.00	0.98 25.00	2.91 74.00	2.75 70.00	3.46 88.00	2.75 70.00	1.36 34.50	0.55 14.00	0.55 14	7.87 200	6.30 160
1 1/4"	1 1/4 11 1/2	1 5/8 12	0.79 20.00	0.79 20.00	4.45 113.00	0.98 25.00	0.98 25.00	2.91 74.00	2.75 70.00	3.46 88.00	2.75 70.00	1.36 34.50	0.55 14.00	0.55 14	7.87 200	6.30 160

Each cell: inches (top) and mm (bottom); d = thread size / threads per inch. For reference only.

WHAT THE LETTERS MEAN

d port thread size · i thread depth · LW port bore · L overall length · I body length · B body width · H height to top of handle hub · h body height · m bore centreline to base · V handle hub above body · SW stem across flats · K1 · K2 handle reach, two positions

SPECIFICATIONS & TIPS

RATING · CONNECTIONS

10,000 psi (689 bar) · soft seat · hard-chrome ball · NPT, BSP, BSPT, ISO, SW, SAE

NPT OR SAE?

NPT seals on the tapered thread with sealant; SAE seals on an O-ring — no sealant in the system.

BEST FOR

Hydraulic power units, test benches and mobile hydraulics.

INSTALLATION, CARE & ORDERING

A few simple habits keep a ball valve **leak-free and easy to turn** for years.

INSTALLING



Check before you fit

Confirm size, material and pressure rating on the valve marking against your specification. Flush the line of chips and debris first.



Hold the body, not the handle

Back up with a wrench on the body or end flats when tightening connections. Never use the handle as a lever.



Threaded ends

Apply sealant or tape to the male thread only, and keep it off the first thread so none enters the valve.



Tube ends

Cut tubing square, deburr, insert fully and follow the tube fitting make-up procedure. Keep ferrule sets together.

OPERATING & CARE



Fully open or fully closed

Ball valves are for on/off and switching duty. Running one part-open to throttle flow erodes the seats and causes leaks.



Read the handle

On two-way valves, handle in line with the pipe = open. On 3-way valves the handle points to the flow path.



Stem weeping?

Snug the gland nut a small amount — the stem packing is adjustable with the valve still in line.



Safety first

Depressurise and drain the line before loosening any end connection or packing, and never exceed the rated pressure.

ORDERING CHECKLIST



Series or part number

From the tables in this catalog



Size & end connection

NPT / BSP / BSPT, tube OD (inch or metric), SW or SAE



Body & seat material

Or tell us the fluid and we will recommend



Pressure & temperature

Normal and maximum operating conditions



Documentation

Material test reports, NACE MR0175 if required



Quantity & delivery

Quantities, required date and ship-to location

NEED A HAND CHOOSING?

Tell us the application — we'll match the valve.

Send your fluid, pressure, temperature and connection details and our team will recommend a series, confirm materials and quote.



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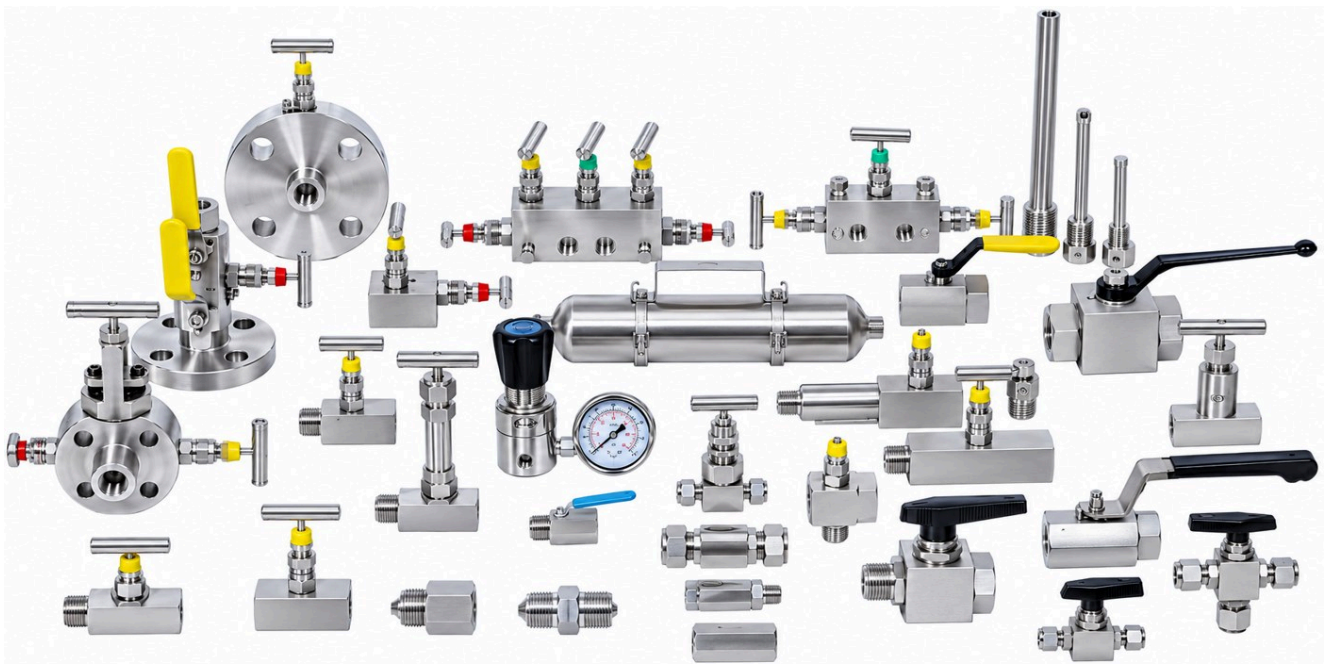


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INSTRUMENTATION VALVES · FITTINGS · ACCESSORIES

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